

AKSHATH PRASAD

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EDUCATION

Indian Institute of Technology Delhi

M.Tech, Computer Technology | CGPA: 9.25/10 | Department Rank: 2

2025 – Expected 2027

Delhi, India

Indian Institute of Information Technology, Kota

B.Tech, Computer Science and Engineering | CGPA: 8.63/10

2021 – 2025

Kota, India

EXPERIENCE

Infosys Limited (ICETS/STG)

Jun 2026 – Jul 2026

Summer Intern

- Engineered a heterogeneous Graph Attention Network (GAT) for fraud detection on the IEEE-CIS graph (3.5% fraud rate), generating reusable 8-dim transaction embeddings; achieved ROC-AUC 0.873 and $4\times$ PR-AUC improvement over linear baselines.
- Designed a 12-qubit Variational Quantum Classifier (VQC) using CZ-gate entanglement (ROC-AUC 0.81); applied SHAP/QSHAP to prune $\sim 31\%$ of VQC parameters without sacrificing accuracy.

Sigfyn Financial Services Pvt. Ltd.

Feb 2025 – Jul 2025

Full Stack Intern

- Built reusable React Native components from Figma designs and fixed production bugs across the React Native frontend and Python/Azure Functions backend.
- Integrated Face ID/Touch ID authentication and Cosmos DB-backed data flows.

RESEARCH EXPERIENCE

Adaptive Noising in Mixed Space for Efficient Diffusion-Based Molecular Generation

Under review, ICLR 2027

M.Tech Thesis, Advisor: Prof. Sandeep Kumar, IIT Delhi

- Designed token-wise adaptive noising in a coupled continuous-discrete space using Gaussian noising and discrete masking, adapting corruption strength to per-token denoising difficulty.
- Developed trajectory-aware reduced-step decoding achieving up to $170\times$ inference speedup with minimal quality loss; outperformed diffusion baselines on ChEBI-20 and M3-20M.

BEM-FIT: Graph-Aware Molecular Tokenizer

Under review, Nature Communications

Advisor: Prof. Sandeep Kumar, IIT Delhi

Apr 2026

- Designed an MDL-based SMILES tokenizer combining Shannon entropy, graph mutual information, and edge entropy; built $O(1)$ incremental merge scoring over a 1.5M-molecule ZINC20 corpus.
- Pretrained MLM encoders on 9.96M molecules across 7 tokenizers, improving all primary MoleculeNet metrics over baselines (ROC-AUC +0.026, RMSE -0.11).

PROJECTS

Local Hybrid Semantic Document Search Engine [GitHub]

Aug 2026

- Built hybrid FAISS + BM25 retrieval fused with Reciprocal Rank Fusion (RRF); implemented multi-format ingestion, OCR, multiprocessing, and event-driven index hot-reloading.
- Exposed a RESTful FastAPI backend achieving 41ms p95 latency; Dockerized for portable deployment.

eBPF-Based Predictive Page Prefetcher with Online Markov Modeling [GitHub]

Aug 2026

- Built a Linux page prefetcher using eBPF tracepoints and online order-1/order-2 Markov models to predict file access patterns and trigger prefetching in real time.
- Benchmarked against Linux readahead on a strided 200MB workload, achieving near-zero page-cache misses and $\sim 11\%$ lower end-to-end I/O time.

tinytorch: Deep Learning Library with Custom Autograd Engine [GitHub]

Apr 2026

- Built a pure-NumPy deep learning library with dynamic autograd ($< 10^{-6}$ numerical error vs. PyTorch), implementing Linear, Conv2d, LayerNorm, AdamW, and SGD with a PyTorch-style API.
- Implemented GPT-2 with causal self-attention, KV-caching, and int8 quantization; ran real Hugging Face weights and trained a custom MLP to 98.2% MNIST accuracy.

Multi-Cloud Serverless ML Deployment Platform

May 2026

- Built a FastAPI platform deploying ONNX models via Celery/Redis with JWT authentication and Cloudflare R2; deployed on Azure Container Apps with scale-to-zero compute.
- Added AWS/Azure deployment adapters and Kubernetes HPA autoscaling (2–10 replicas); instrumented the platform with Prometheus.

32-Bit ALU with High-Speed Multiplier and Divider

Oct 2025

- Designed and verified a 32-bit Verilog ALU; implemented Radix-4 Booth multiplication, Dadda-tree reduction, Kogge-Stone addition, non-restoring division, and barrel shifting, with synthesis-based timing, area, and power analysis.

NTP-Synchronized Speaking Clock on FreeRTOS/lwIP

Apr 2026

- Engineered a QEMU ARM Cortex-M3 system with four concurrent FreeRTOS tasks, semaphore/message-queue synchronization, and an SNTP client over lwIP with DHCP/static-IP fallback and UTC-to-IST conversion.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, SQL, Bash

ML/AI: PyTorch, Hugging Face, NumPy, scikit-learn, Transformers, Diffusion Models, Graph Neural Networks, LLMs/NLP, FAISS, SHAP

Software/Systems: Linux, eBPF, FastAPI, REST APIs, React Native, Celery, Concurrency, Multiprocessing, OS/Computer Architecture, FreeRTOS, lwIP

Cloud/Data: Docker, Kubernetes, AWS, Azure, Prometheus, Redis, PostgreSQL, MongoDB, MySQL, Git